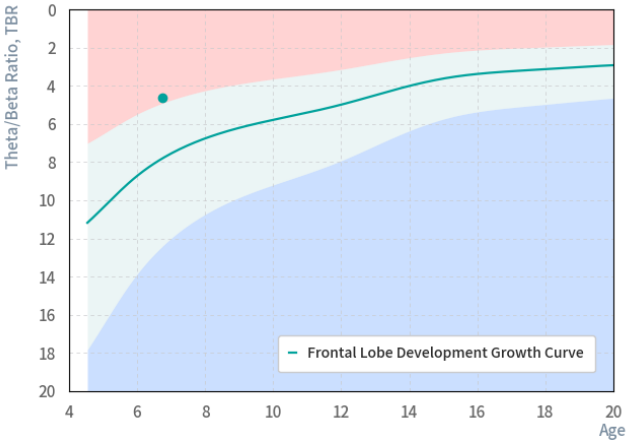
※The summary report provides analysis results based on the information at the time of measurement. If you change customer's personal information after completion of analysis, please reanalyze through iSyncBrain and check the result.

Brain Function Mapping
(With Brain Development Indicators)

Client name	Sex	Date of Birth(age)	EEG acq. date	Eyes Closed/Open
Isaac Franca	Male	2018.11.06 (6Y)	2025.08.04	Eyes Open

1 Frontal Lobe Development Indicators

Frontal Lobe Development Graph



Frontal Lobe Development Indicator Percentile	86.9%ile	Your Indicator	4.6

* This indicator provides the most accurate results when the eyes are open.

Frontal lobe development can be assessed through the Theta/Beta Ratio (TBR).

TBR represents the ratio of theta waves (4-8Hz) to beta waves (13-21Hz) in brainwaves. The FDA has approved the use of TBR as a reference for diagnosing ADHD in pediatric populations. TBR rapidly decreases with age during childhood and adolescence, maintaining a range of 2-3 in adulthood. Higher TBR values in children and adolescents compared to age-matched peers may indicate relative immaturity of the frontal lobe.

If the TBR is excessively higher than the healthy range when the eyes are open, developmental issues such as ADHD or autism spectrum disorders may be suspected.

2 Information Processing Speed Indicator

Information Processing Speed Graph

Information Processing Speed Indicator Percentile	-%ile	Your Indicator	- Hz

* This indicator provides the most appropriate results when the eyes are closed.

The Information Processing Speed Indicator can be determined from the Occipital Peak Frequency, which is the alpha frequency in the occipital lobe.

When eyes are closed, the frequency of alpha waves generated in the occipital region increases significantly with age during childhood and adolescence, stabilizing within the 10 Hz range in adulthood. Lower alpha frequencies compared to age-matched peers during childhood and adolescence may suggest slower information processing in the brain.

If the alpha frequency is excessively lower than the healthy range when the eyes are closed condition , developmental issues in the brain's information processing abilities, ranging from sensation to perception and cognition, may be possible.

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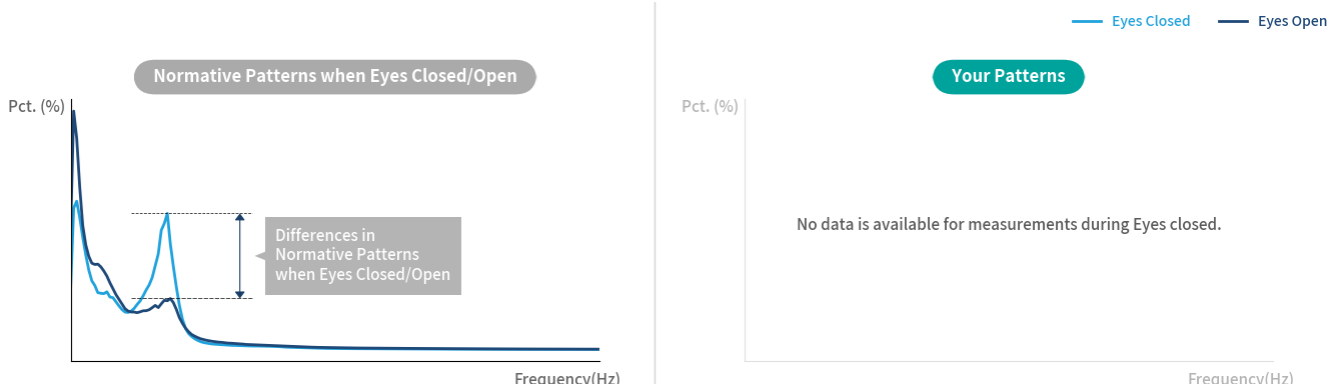
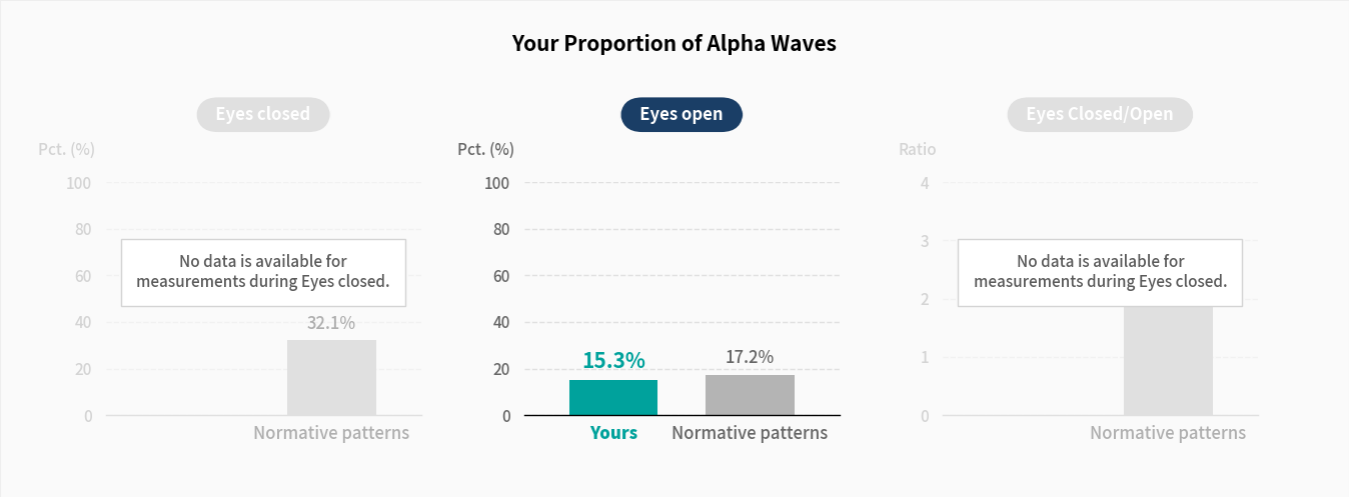
3 Alpha Wave Reactivity Indicator

The Alpha Wave Reactivity Indicator is determined by the proportion of Relative Alpha Power.

It signifies the proportion of Alpha band in your occipital brainwaves. In childhood and adolescence, the proportion of Alpha waves increases with age, reaching about 40% of total brainwaves when eyes are closed and around 20% when eyes are open in adulthood. The proportion is approximately twice as much when eyes are closed compared to when they are open.

If the proportion of Alpha waves in childhood and adolescence is excessively lower compared to peers, it may indicate an immature age-related information processing capacity. Additionally, if the proportion of Alpha waves does not decrease more than twice when eyes are open compared to closed, it could also pose challenges to information processing capabilities.





Analysis Comments

Your Alpha Wave Reactivity Analysis results indicate that, compared to the same age group, the alpha wave proportion is lower when eyes are closed but falls within the average range when eyes are open.

The lower Alpha Wave proportion when eyes are closed suggests a state similar to when the eyes are open, indicating a partial wakefulness.

* This indicator provides the most appropriate results when measuring both eyes closed and eyes open conditions.

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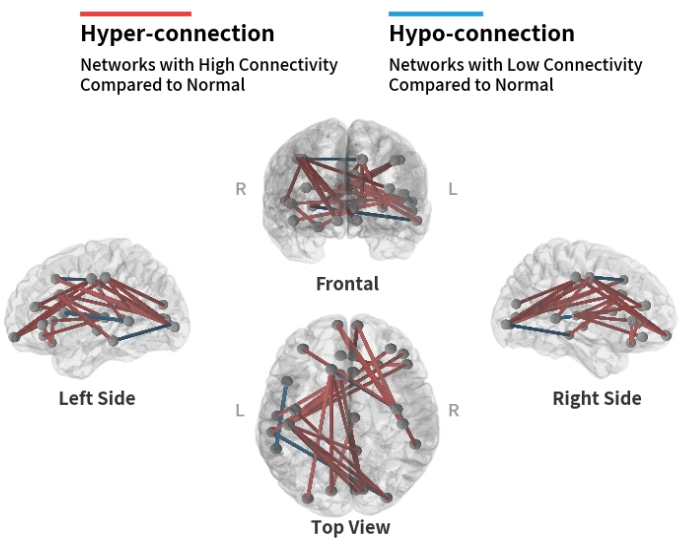
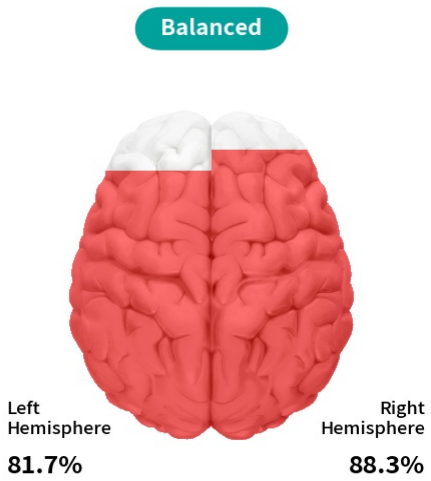
4 Brain Balance

Brain Balance can be assessed through the connectivity of the brain.

After identifying the connectivity between each of the 68 anatomical brain regions and determining lateralization, i.e., whether the left or right hemisphere is more activated, we can assess the Alpha wave connectivity pattern. Quantifying the degree of activation compared to the norm allows us to categorize the lateralization as left-brained or right-brained. Higher values suggest stronger connectivity, while low connectivity in both regions predicts impairment.

If the left hemisphere is dominant, language abilities and logical reasoning may be well developed, while dominance of the right hemisphere may lead to superior emotional abilities and enhanced spatial processing. Reduced Alpha band connectivity between hemispheres may raise suspicions of autism spectrum disorder, while excessively increased connectivity at high Alpha bands may suggest ADHD.





Analysis Comments

Your Brain Balance Analysis results show 81.7% in the left hemisphere and 88.3% in the right hemisphere, indicating a 'Balanced' dominance.

* This indicator provides the most appropriate results when the eyes are closed.

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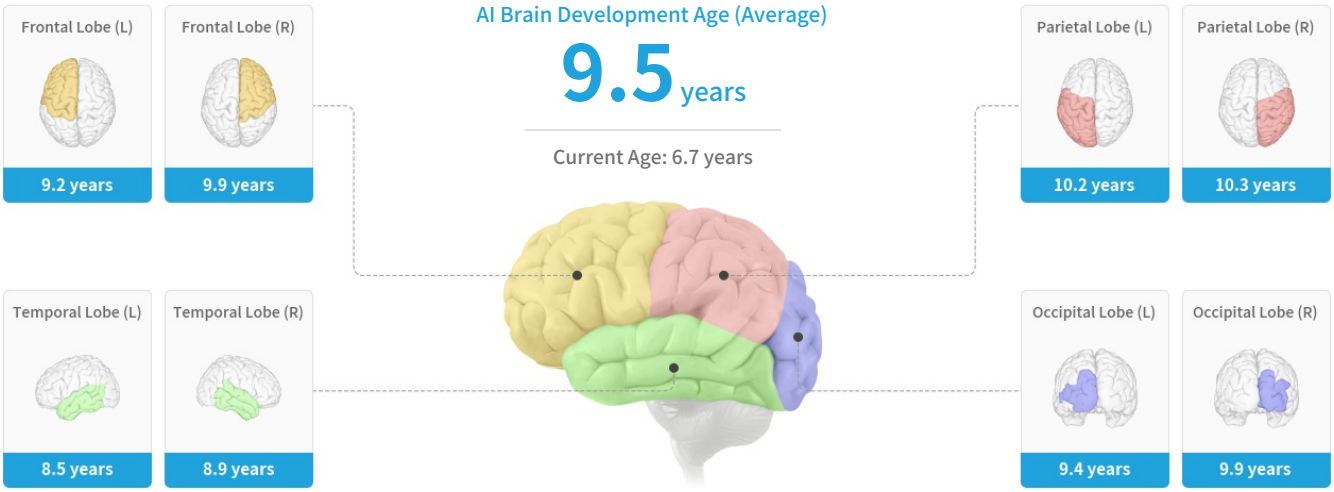
Isaac Franca	Male	2018.11.06 (6Y)	2025.08.04	Eyes Open

5 AI Brain Development Age

Using normative data, AI can predict the development age of 8 brain regions. These include four areas in each hemisphere: the frontal lobe, parietal lobe, temporal lobe, and occipital lobe.

The left hemisphere processes sequential information, such as language and logic, while the right hemisphere handles simultaneous information, like emotions and spatial processing. The occipital and temporal lobes receive external information, which is subsequently integrated in the parietal lobe before being processed or stored in the frontal and temporal lobes. Based on the AI analysis results for each region, a comparison can be drawn between the customer's biological age and their brain development age.



* This indicator provides the most accurate results when the eyes are open.

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